1 **(a)** 2 hours 15 minutes **[B1]**

(b) Adds 2 hours 35 minutes to 16 45 (e.g., 18 80) **[M1]**

19 20 **[A1]**

[Total Marks: 3]

2 Arrival time in London local time = 21 : 45 + 6 hours 50 minutes **[M1]**

04 : 35 (next day) **[M1]**

Dubai local time = 04 : 35 + 4 hours = 08 : 35 **[A1]**

[Total Marks: 3]

3 Flight arrival time in Dubai local time: 14 30 + 14 h 15 min = 28 45 (or 04 45 next day) **[M1]**

Adds time zone difference: 28 45 + 7 hours (or 04 45 + 7 hours) **[M1]**

11 45 **[A1]**

[Total Marks: 3]

4 **(a)** 11 : 03 − 08 : 15 **[M1]**

2 hours 48 minutes **[A1]**

(b) 48 minutes $= \frac{48}{60}$ hours = 0.8 hours **[M1]**

Average speed $= \frac{420}{2.8}$ **[M1]**

150 **[A1]**

[Total Marks: 5]

5 **(a)** 920 ÷ 24 **[M1]**

38.333... hours **[A1]**

38 hours 20 minutes **[A1]**

(b) Adds their time from part (a) to Wednesday 17 50 **[M1]**

Wednesday 17 50 + 20 minutes = 18 10 **[M1]**

Wednesday 18 10 + 38 hours = Thursday 18 10 + 14 hours **[M1]**

Day: Friday **[A1]**

Time: 08 10 **[A1]**

[Total Marks: 7]

6 **(a)** Minutes in a year = 365 × 24 × 60 **[M1]**

525600 minutes **[M1]**

Number of orbits $= \dfrac{525600}{92} = 5713.043\ldots$ **[M1]**

5713 **[A1]**

(b) (i) $\dfrac{11440}{880} = 13$ **[B1]**

(ii) Time difference $= 9 - (-4) = 13$ hours (Q is 13 hours ahead) **[M1]**

Departure time in City Q's local time $= 22:45$ (Monday) $+13$ hours $= 11:45$ (Tuesday) **[M1]**

Arrival time in City Q's local time $= 11:45$ (Tuesday) $+13$ hours (flight time) $= 00:45$ (Wednesday) **[A1]**

[Total Marks: 8]